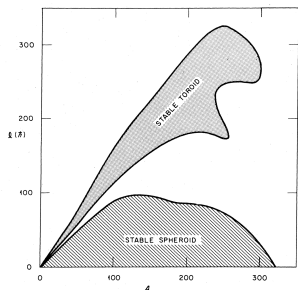


The Toroidal Nucleus Search in ^{28}Si at TAMU Cyclotron Institute

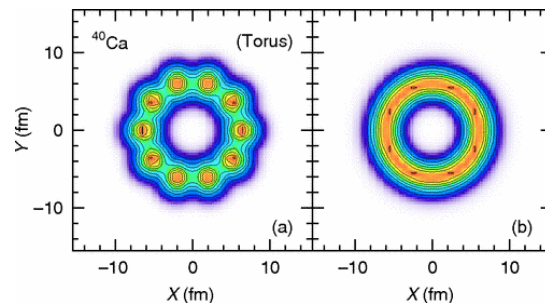
Bryan M Harvey, September 2025

An Incomplete Theoretical Background (Pre Cao et al.)



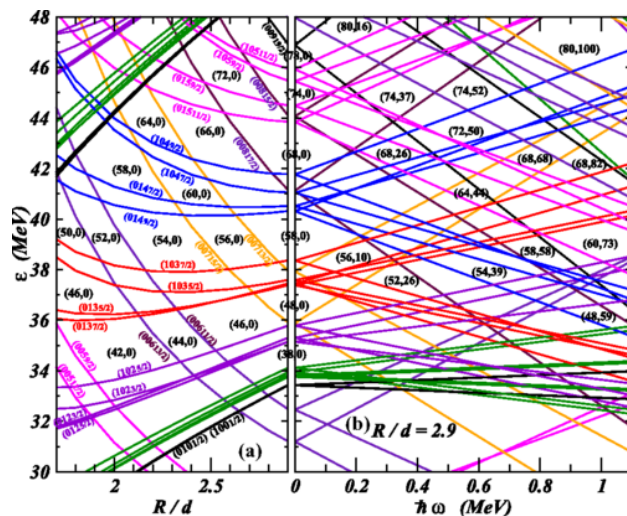
Phys. Rev. C **17**, 331 (1978)

- Bulk regions of I - A space give theoretical toroidal stability



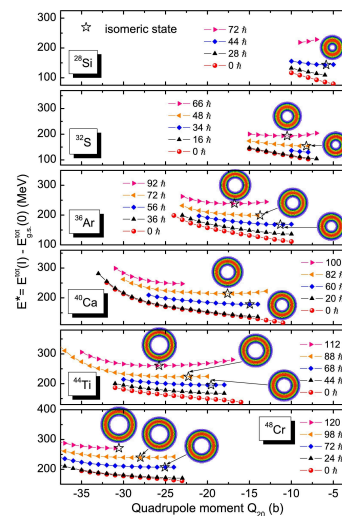
Phys. Rev. Lett. **109**, 1103 (2012)

- HF suggests $^{40}\text{Ca}^*$ state at 170 MeV



Phys. Rev. C **98**, 034316 (2018)

- Single particle excitation energies explored as a function of deformation



Phys. Lett. B **738**, 401 (2014)

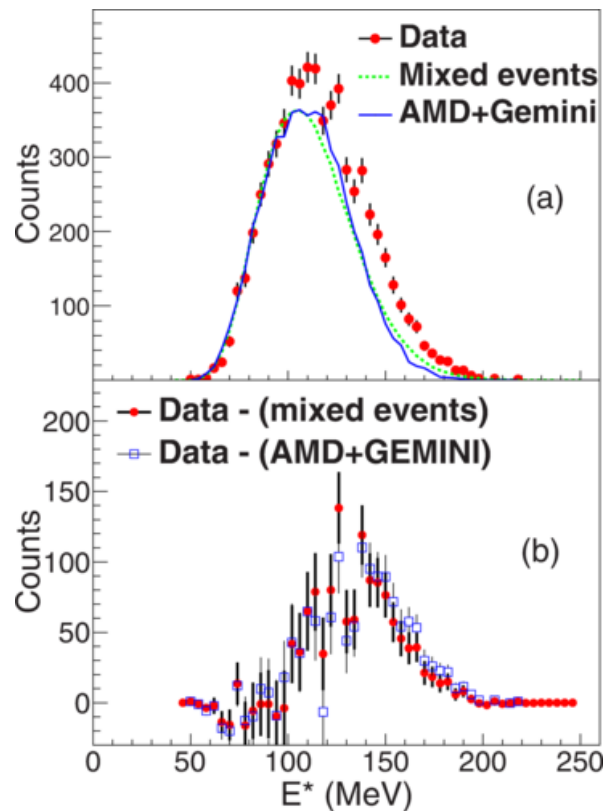
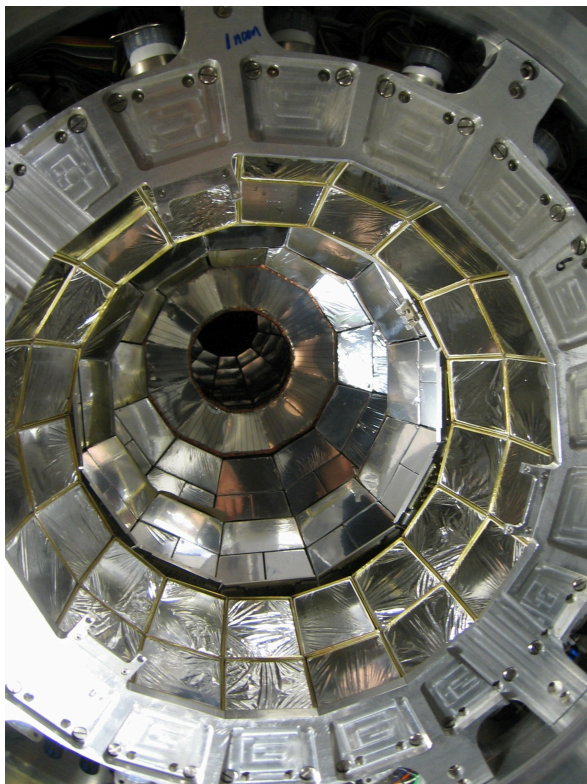
- Cranked Skyrme-HFB
- **Specific E^* and I predictions**
- For ^{28}Si :

$$E^* = 143.18 \text{ MeV}$$

$$I = 44\hbar$$

Experimental Background - NIMROD (Cao *et al.*)

Neutron Ion Multidetector for
Reaction Oriented Dynamics



Phys. Rev. C **99**, 014606 (2019)

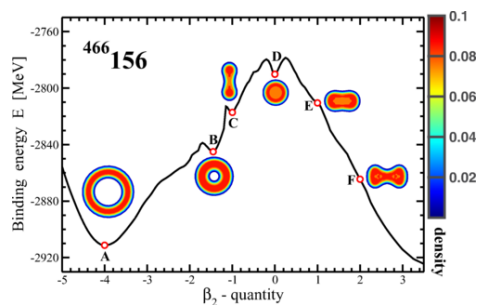
- 2009 data as part of a much larger α -conjugate study
- $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV / u

-
- $E^* = \sum_i K_{i,\text{c.o.m.}} - Q$
 - $E^* = 114, 126, \mathbf{138}$ MeV structures observed

- Background subtraction

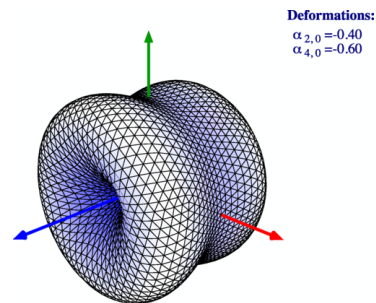
-
- Limited E^* Resolution
 - ~9.4 MeV FWHM at 138 MeV
 - Meas. widths $\approx$ Detector Res.
 - Limited Sample Size
 - ~6.5k 7α events

A Very Incomplete Theoretical Background (Post Cao et al.)



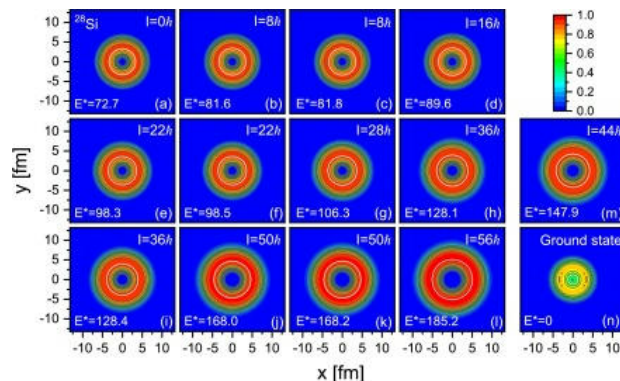
Phys. Rev. C **103**, 034323 (2021)

- Exotic hyper heavy deformation, including toroids



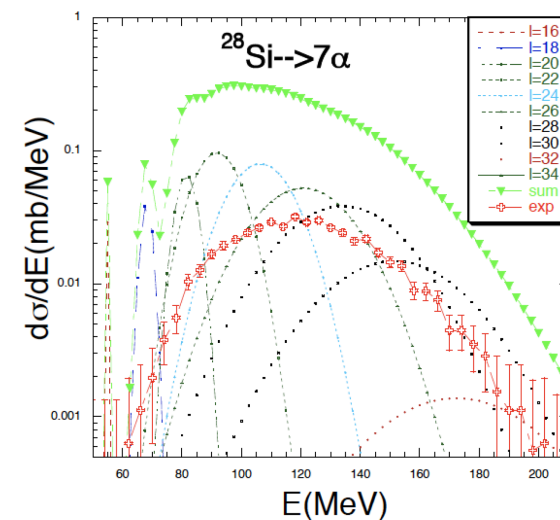
Phys. Rev. C **103**, 054311 (2021)

- Mean-field calculations to predict toroidal geometries



Nuc. Phys. A **996**, 121696 (2020)

- CDFT Calculations
- Many predictions in ^{28}Si

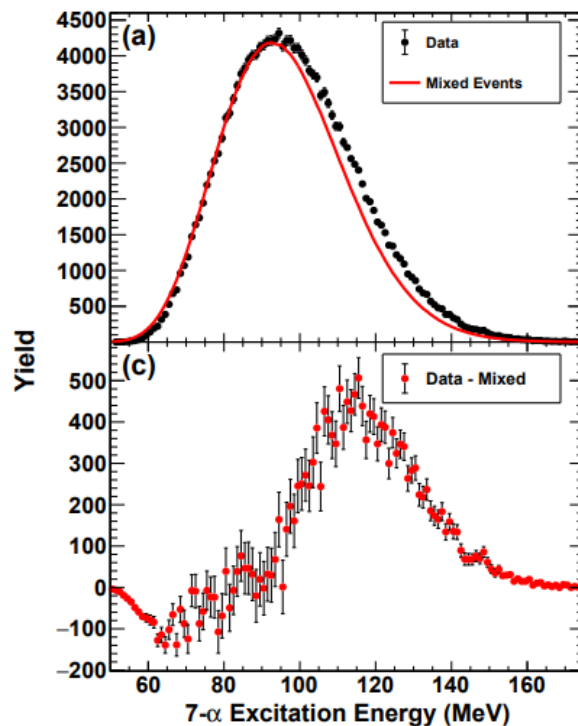
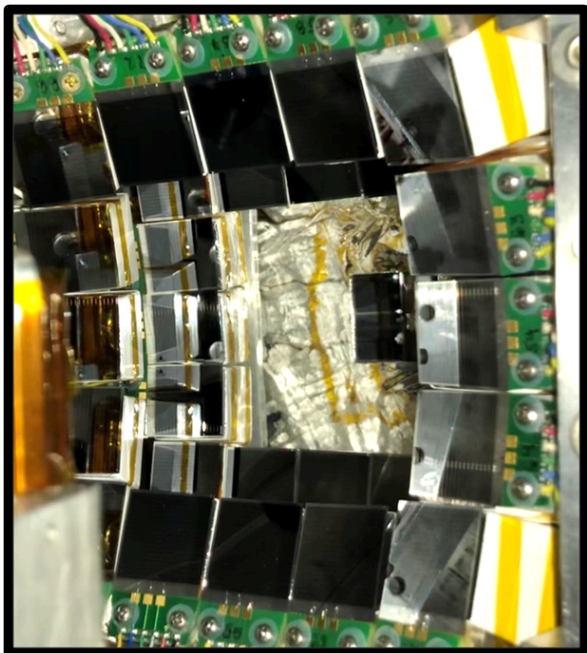


Symmetry **13**, 1777 (2021)

- Hybrid α -Cluster (H α C) model
- Breakup probabilities assessed at various l and E^*

Experimental Background - FAUST (Hannaman *et al.*)

Forward Array
Using Silicon Technology



Phys. Rev. C **109**, 054615 (2024)

- Targetted search for high E^* states in ^{28}Si

—

- ~2.5 MeV FWHM at 138 MeV ($\downarrow$ 3.75x)

- ~186k 7α events ($\uparrow$ 29x)

—

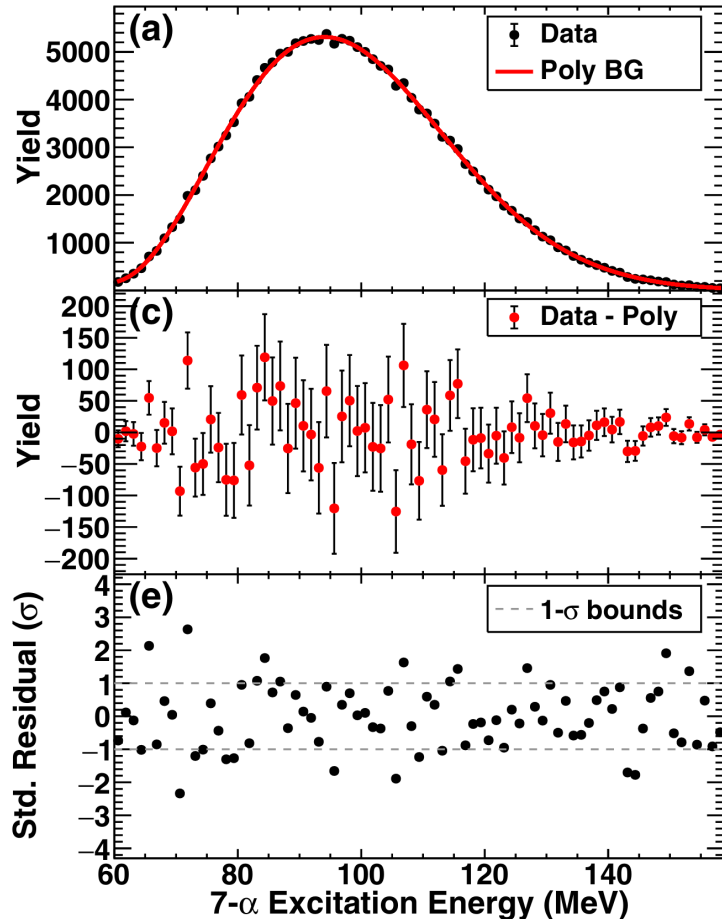
No strong narrow structure at 114, 126, or 138 MeV

-
- No accurate background estimate

✗ Mixed Events

✗ MD Sims

Polynomial Fit



- The *only* purpose of this fit is to demonstrate the entire spectrum is consistent with some broad, ‘featureless’ description.
 - ‘featureless’ - no prominent oscillations on scales below 10 MeV
- The fit is good:
 - Residuals shown
 - $\chi^2_V = 1.01$
- Only claim that “*no strong evidence was found for statistically significant resonant state yield in the seven α -particle channel*”.

New Attempts to Extract Peaks

Introduction to Common Themes

When is a feature consistent with another?

Absolute difference?

Statistical significance?

Measurement Error?

'Close to', 'reproduces', etc.

What is the background?

Shape

Size

Confidence

Importance

Use of Polynomial

What does the fit represent?

What conclusions can be drawn from the fit?

Quality of Calibration

Accuracy

Stability

General Statistics

Independence of samples

Significance of deviations in spectra

Systematics of comparisons to experiment

Gaussian Mixture Model (Depastas *et al.*)

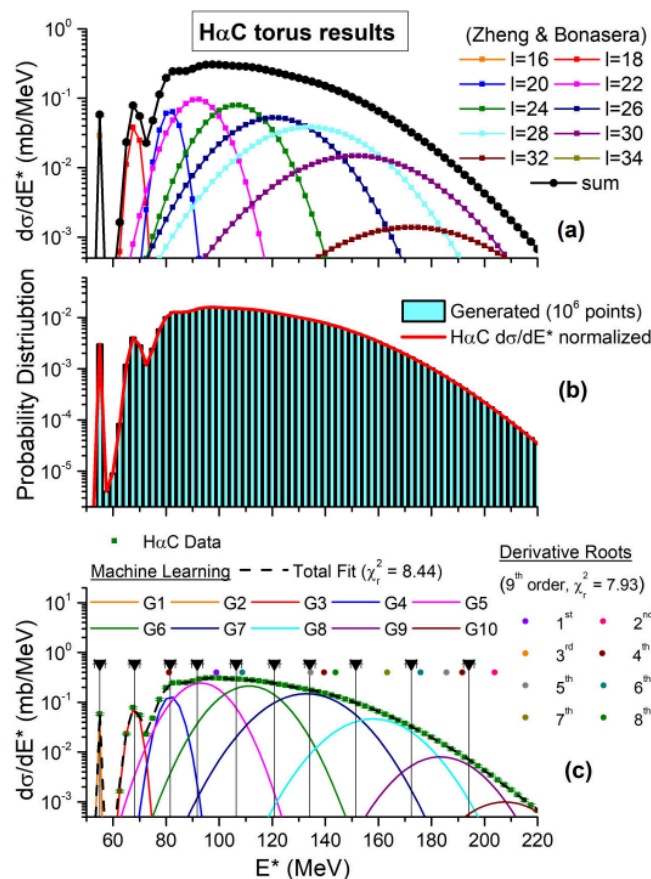
- Spectrum is binned and bins are assigned uncertainty

- Binned data is sampled $N_p = 10^6$ times

- Point cloud is passed to Gaussian Mixture Model for various N_G

- Select the model that optimizes χ^2_v

- Locations of resultant Gaussians are compared to previous predictions

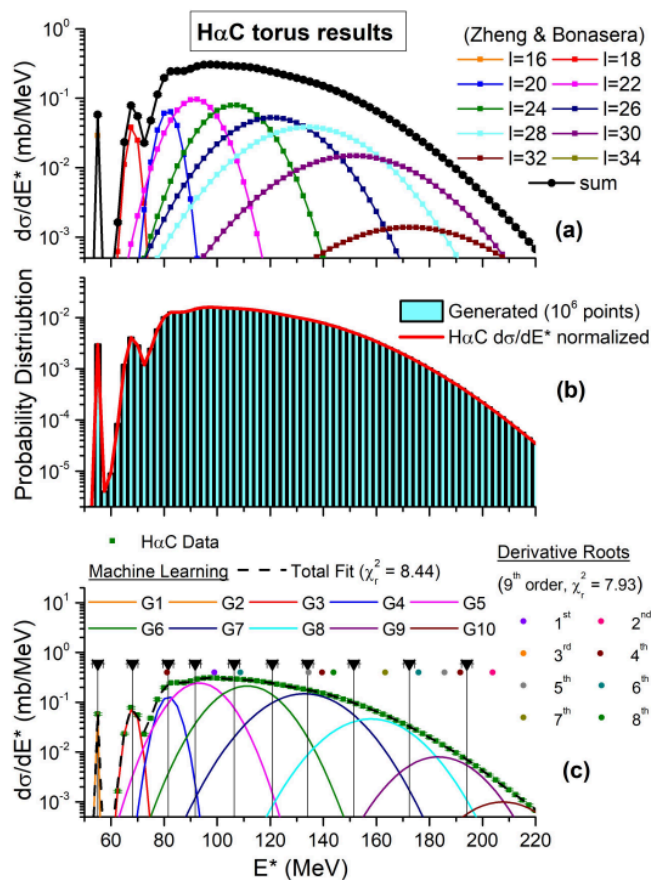


Phys. Rev. C **112**, 014614 (2025)

- Experimental data is already provided as binned with Poisson uncertainties
 - Not shown here
- Model data is calculated at bin centers and bin errors are set to 5%
 - See **RED** in (b)

Gaussian Mixture Model (Depastas *et al.*)

- Spectrum is binned and bins are assigned uncertainty
- **Binned data is sampled $N_p = 10^6$ times**
- Point cloud is passed to Gaussian Mixture Model for various N_G
- Select the model that optimizes χ^2_v
- Locations of resultant Gaussians are compared to previous predictions



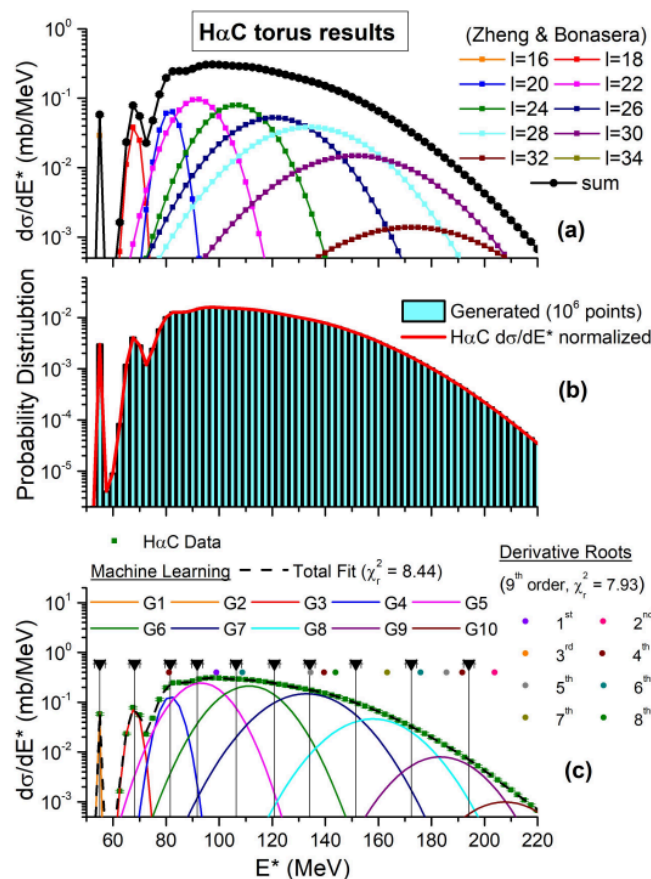
Phys. Rev. C **112**, 014614 (2025)

“From this discrete histogram, we generate $N_p = 10^6$ random points, from a normal distribution around the center of each bin”

- Unspecified width of sampled Gaussian
- See **CYAN** in (b)

Gaussian Mixture Model (Depastas *et al.*)

- Spectrum is binned and bins are assigned uncertainty
- Binned data is sampled $N_p = 10^6$ times
- **Point cloud is passed to Gaussian Mixture Model for various N_G**
- Select the model that optimizes χ^2_v
- Locations of resultant Gaussians are compared to previous predictions



Phys. Rev. C **112**, 014614 (2025)

“A Gaussian mixture model is a probabilistic model that assumes all the data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters”

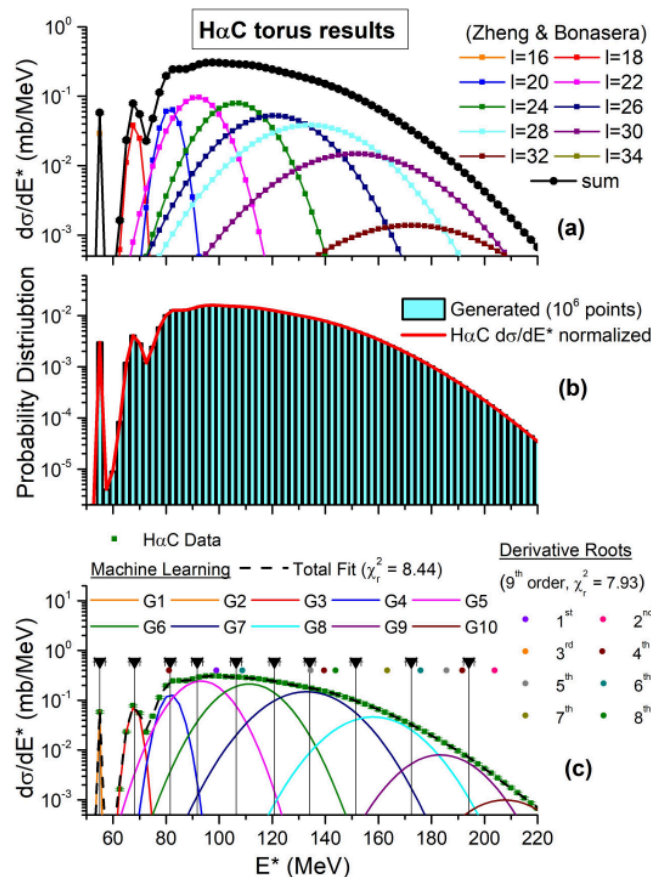
- <https://scikit-learn.org/stable/modules/mixture.html>

-
- Iteratively partition data
 - N_G groups
 - Calculate Gaussians
 - Calculate & maximize $\mathcal{L}$
-

- Final result is similar to multi-Gaussian fit

Gaussian Mixture Model (Depastas *et al.*)

- Spectrum is binned and bins are assigned uncertainty
- Binned data is sampled $N_p = 10^6$ times
- Point cloud is passed to Gaussian Mixture Model for various N_G
- **Select the model that optimizes χ^2_v**
- Locations of resultant Gaussians are compared to previous predictions



Phys. Rev. C **112**, 014614 (2025)

$$\chi_v^2 = \frac{\sum_i \left\{ \frac{1}{\delta y_i} \left[y_i - \sum_{l_G=0}^{N_G-1} w_{l_G} g_{l_G}(E_i^*; \mu_{l_G}, \sigma_{l_G}) \right] \right\}^2}{N_p - 3N_G}$$

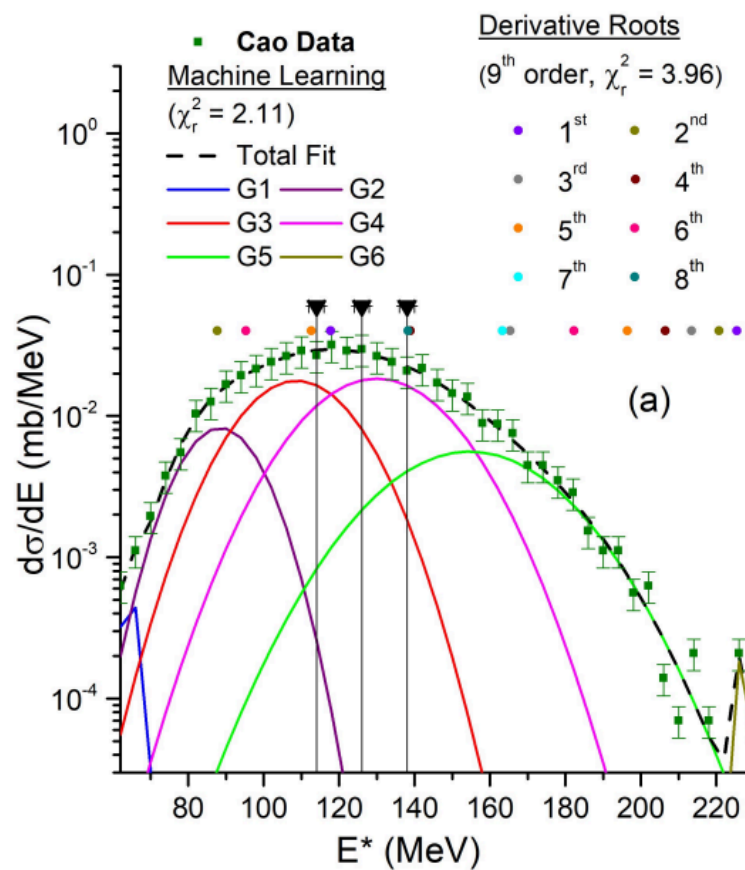
$$N_p = 10^6$$

$$\delta y_i = \begin{cases} \sqrt{y_i} & \text{if Experimental} \\ (5\%)y_i & \text{if Model} \end{cases}$$

- See **various colored** curves in **(c)**, labelled **G1-10**

Application to Experimental Data

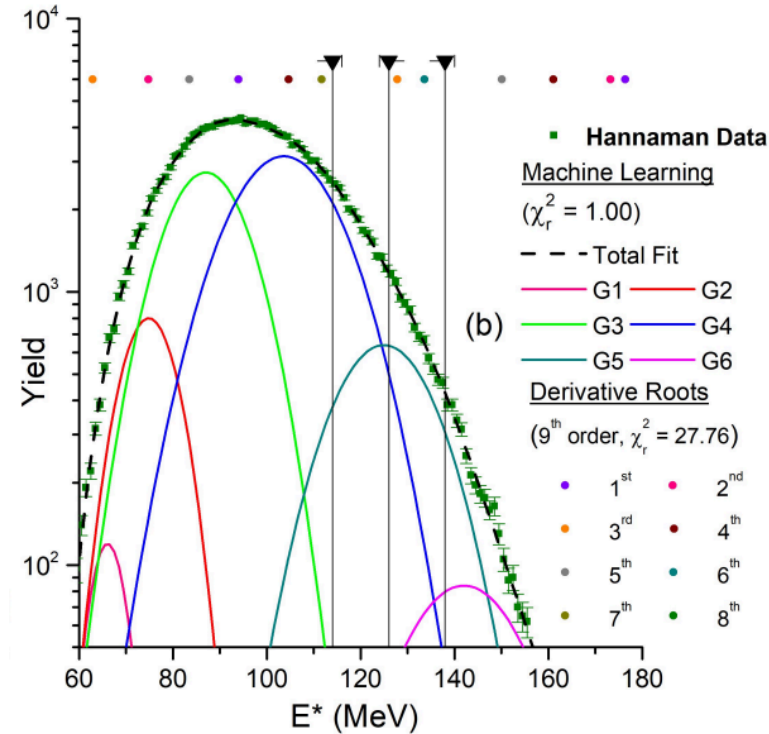
- Spectrum is binned and bins are assigned uncertainty
- Binned data is sampled $N_p = 10^6$ times
- Point cloud is passed to Gaussian Mixture Model for various N_G
- Select the model that optimizes χ^2_v
- **Locations of resultant Gaussians are compared to previous predictions**



- “For the [Cao data] we obtain six peaks which **approximately reproduce** the predicted 114 MeV and 138 MeV toroid resonances (black triangles) in the region of high statistics. The peak at 126 MeV is not reproduced independently, but **belongs to** the wide fourth Gaussian.”
 — §4 par 1
- 114 MeV: ~4 MeV away from G3
- 126 MeV: ~4 MeV away from G4
- 138 MeV: ~8 MeV away from G4

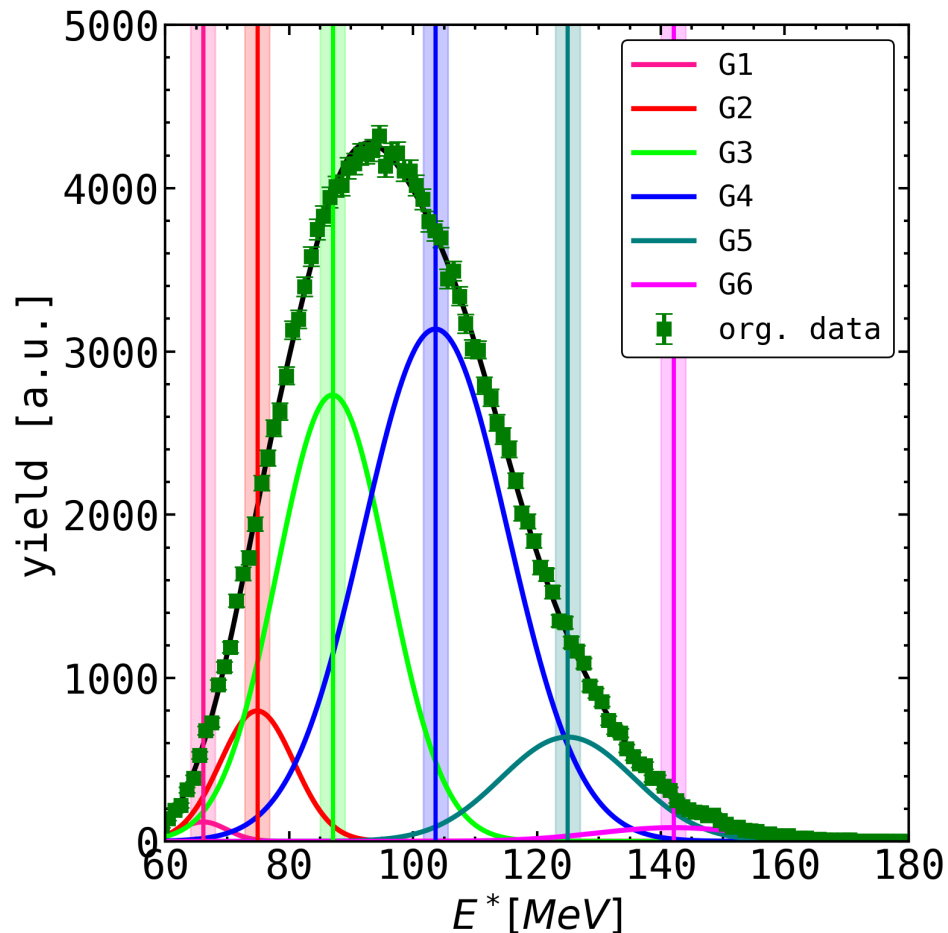
Application to Experimental Data

- Spectrum is binned and bins are assigned uncertainty
- Binned data is sampled $N_p = 10^6$ times
- Point cloud is passed to Gaussian Mixture Model for various N_G
- Select the model that optimizes χ^2_v
- **Locations of resultant Gaussians are compared to previous predictions**



- “Interestingly, the resonances are **close to** the aforementioned found by Cao and collaborators.” — §4 par 1
- 114 MeV: ~10 MeV away from G4
- 126 MeV: ~1 MeV away from G5
- 138 MeV: ~4 MeV away from G6

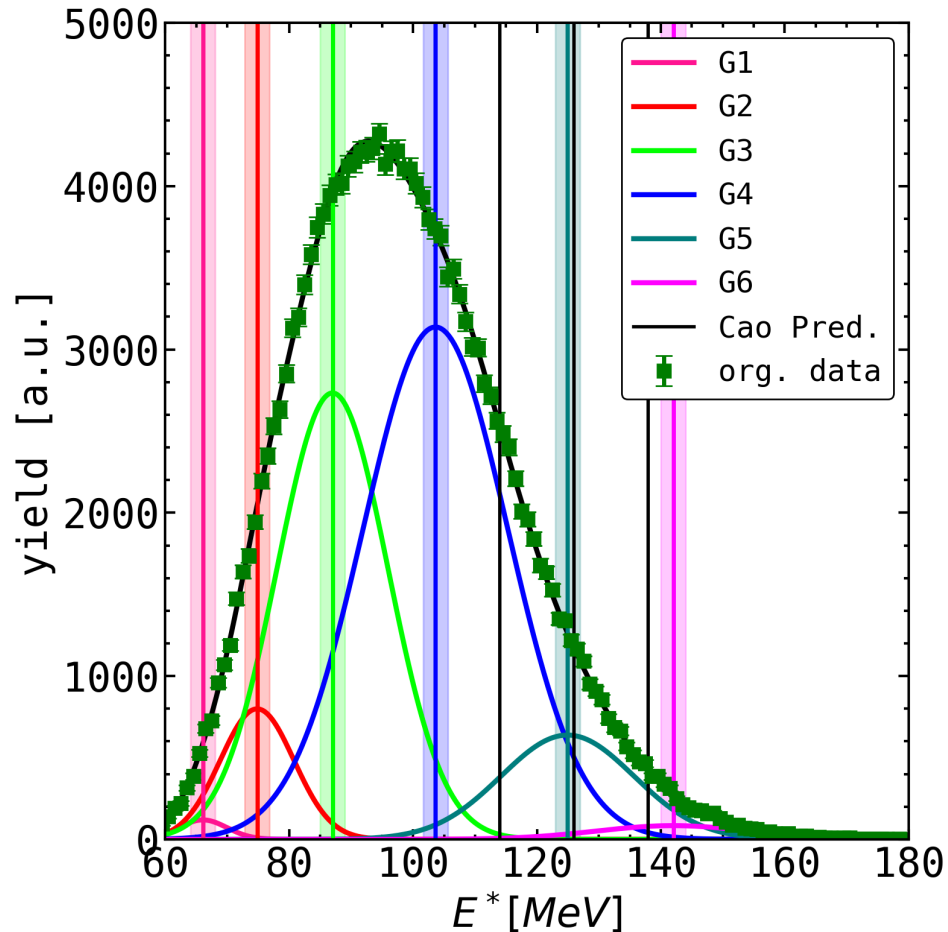
Suppose 2 MeV Windows



- Define “consistency” *before* looking at predictions
- “Any Gaussian that lands within x MeV of a prediction is said to be consistent with that prediction.”
- Ask : “How surprising is it if at least one out of three predictions is reproduced for ± 2 MeV windows?”

$$\begin{aligned}
 P_{\geq 1|3} &= P(n_{\text{repro.}} \geq 1 \mid 3 \text{ pred.}) \\
 &= 1 - \left[P(n_{\text{repro.}} = 0 \mid 1 \text{ pred.}) \right]^3 \\
 &\approx 1 - \left(\frac{120 - 24}{120} \right)^3 \\
 &= \boxed{48.8\%}
 \end{aligned}$$

Add in Cao Predictions



- A statistically unsurprising result:
 - One of the predictions was consistent with one of the Gaussians.

$$P_{=0|3} \approx 51.2\%$$

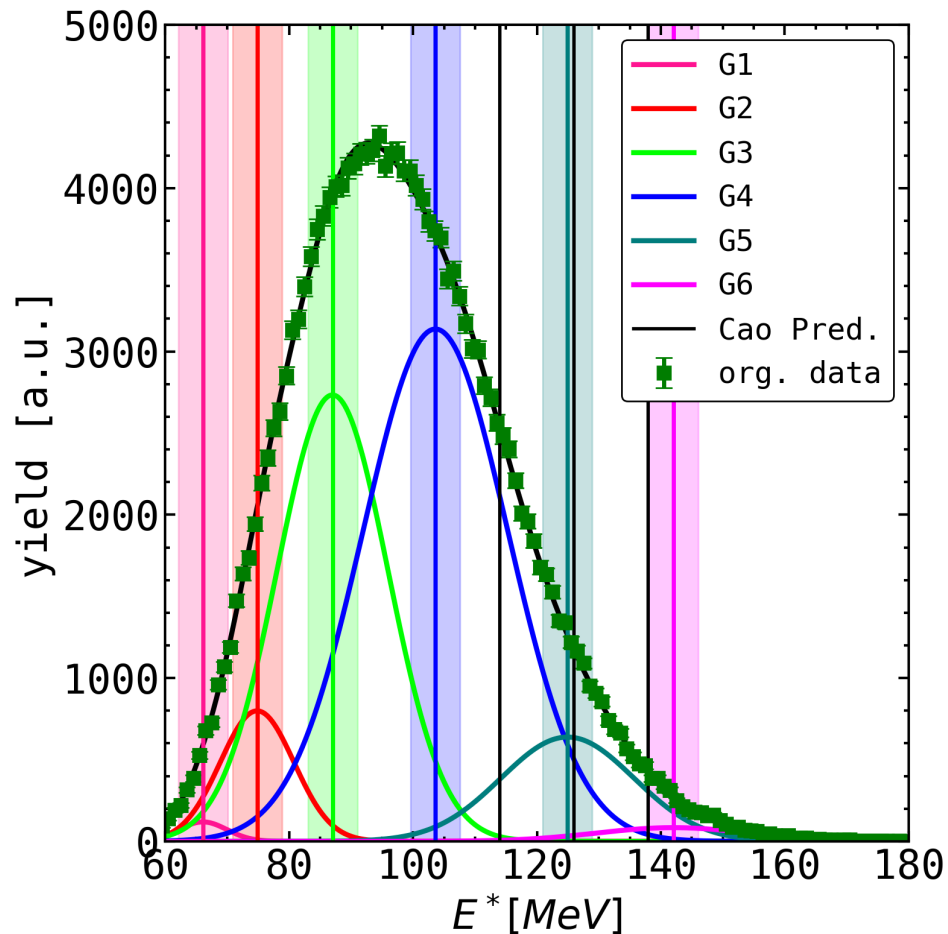
$$P_{=1|3} \approx 38.4\%$$

$$P_{=2|3} \approx 9.6\%$$

$$P_{=3|3} \approx 0.8\%$$

Maybe open the window?

Suppose 4 MeV Windows



- One or two peaks consistent
 - Neither result would be surprising

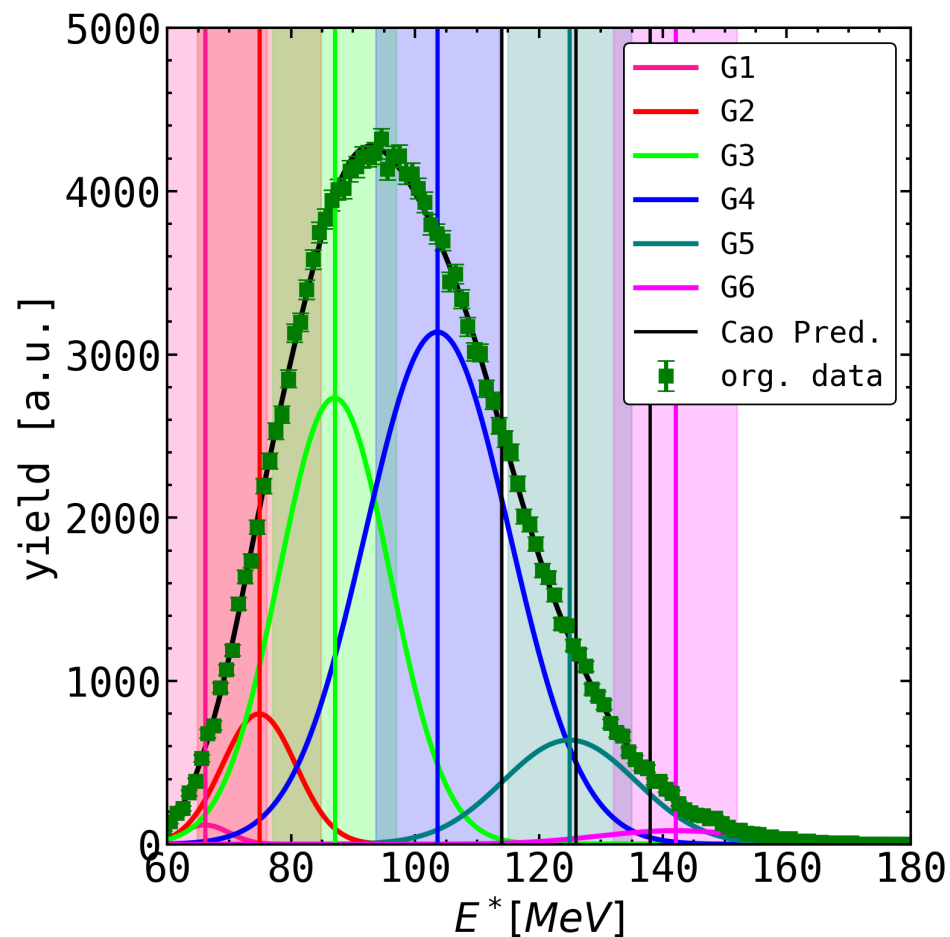
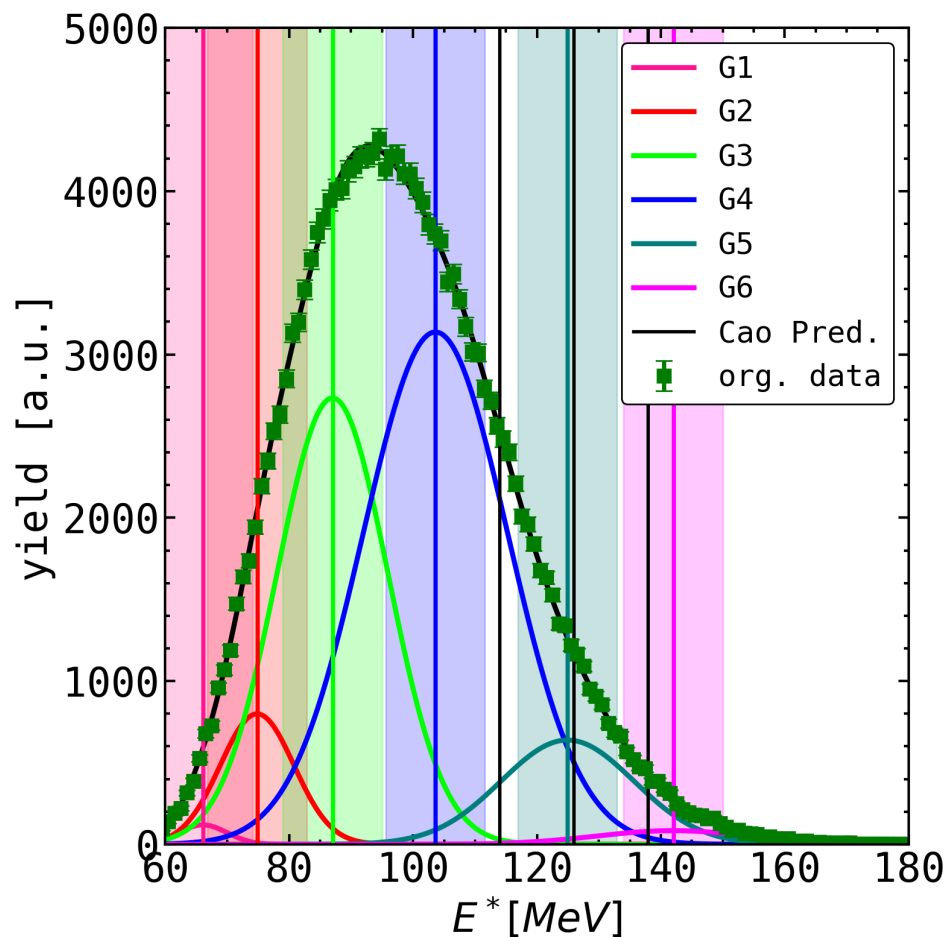
$$P_{=0|3} \approx 21.6\%$$

$$P_{=1|3} \approx 43.2\%$$

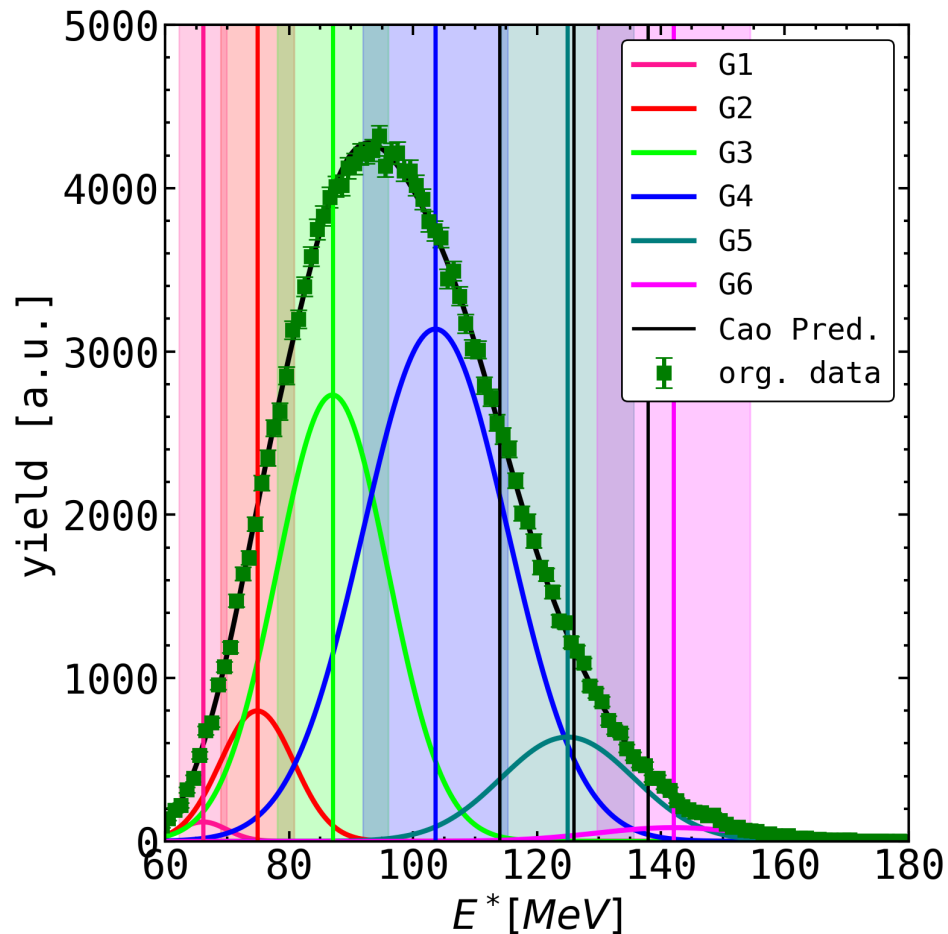
$$P_{=2|3} \approx 28.8\%$$

$$P_{=3|3} \approx 6.4\%$$

Suppose Very Wide Windows



Use HWHM of Each Gaussian

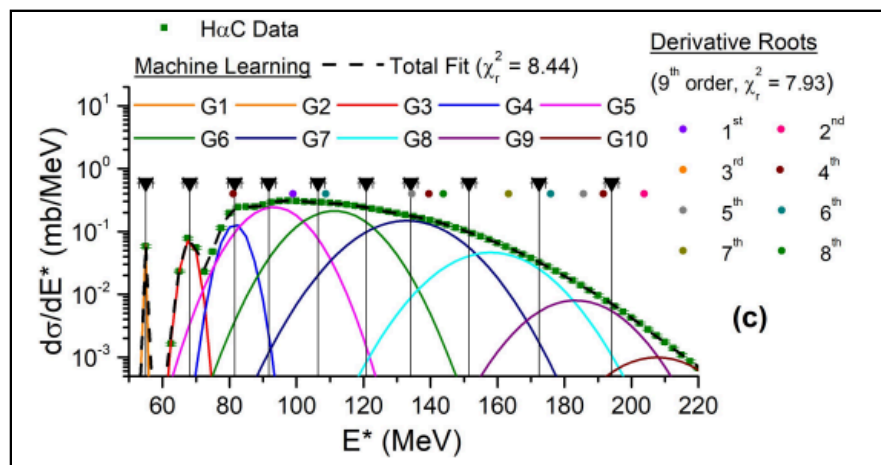
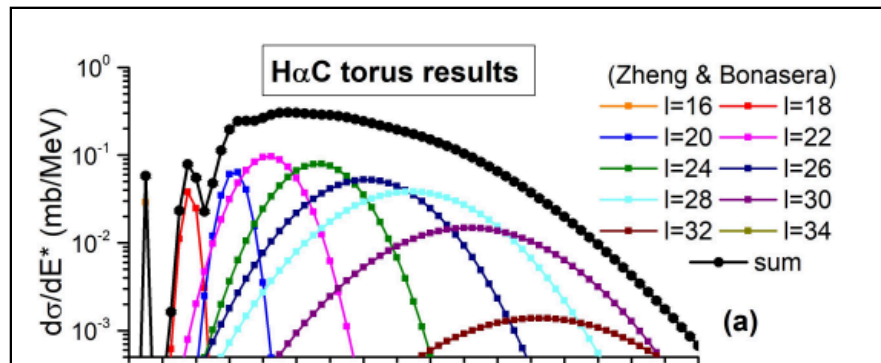


It is challenging to have a definition of consistent/
close/reproducing which simultaneously shows
agreement of the GMM peaks and the Cao
predictions that is statistically significant

The Benchmark

Dependence of Agreement on the Existence of Distinct Features

Using absolute deltas to line up Gaussians within 2 MeV



Correspond to obvious local maxima

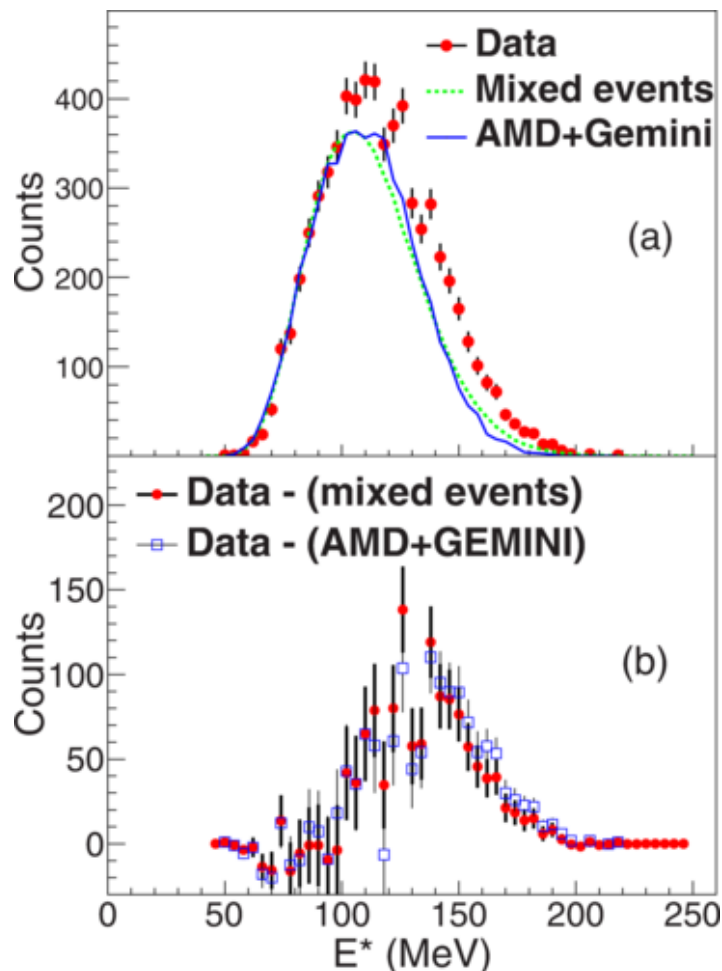
- G1 and G2 : within 1 MeV of $l = 16$
- G3 : within 1 MeV with $l = 18$
- G4 : within 1 MeV with $l = 20$

Do not correspond to obvious local maxima

- G5 : ~ 2 MeV of $l = 22$
- G6 : ~ 4 MeV away from $l = 24$
- G7 : within 1 MeV of $l = 26$
- G8 : ~ 7 MeV away from $l = 30$
- G9 : ~ 10 MeV away from $l = 30$ or $l = 32$
- G10 : ~ 10 MeV away from $l = 32^*$

- Without distinct features, GMM fails to consistently reproduce overlapping contributions

Robustness of Agreement in the Presence of a Background



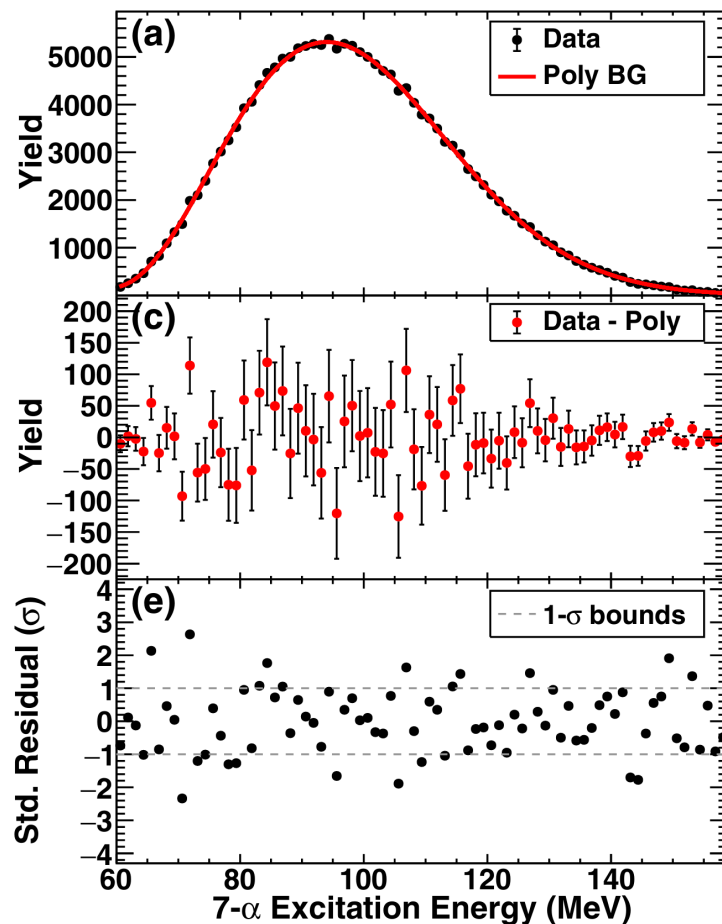
- The background according to MD & mixed:
 - not Gaussian.
 - the dominant feature ($\approx 90\%$) of the distribution.
- No background was included in the benchmark
- GMM assumes the underlying probability distribution is a sum of Gaussians
- The reliability of the GMM technique in the presence of an uncharacterized, but dominant, non-Gaussian background has *not* been demonstrated.

Discussion of the Polynomial Fit

Polynomial Roots

“First, any peak can be fit with a non-linear polynomial, given enough terms. This in turn leads to a loss of possible real peaks in the subtraction” — §2 par 1

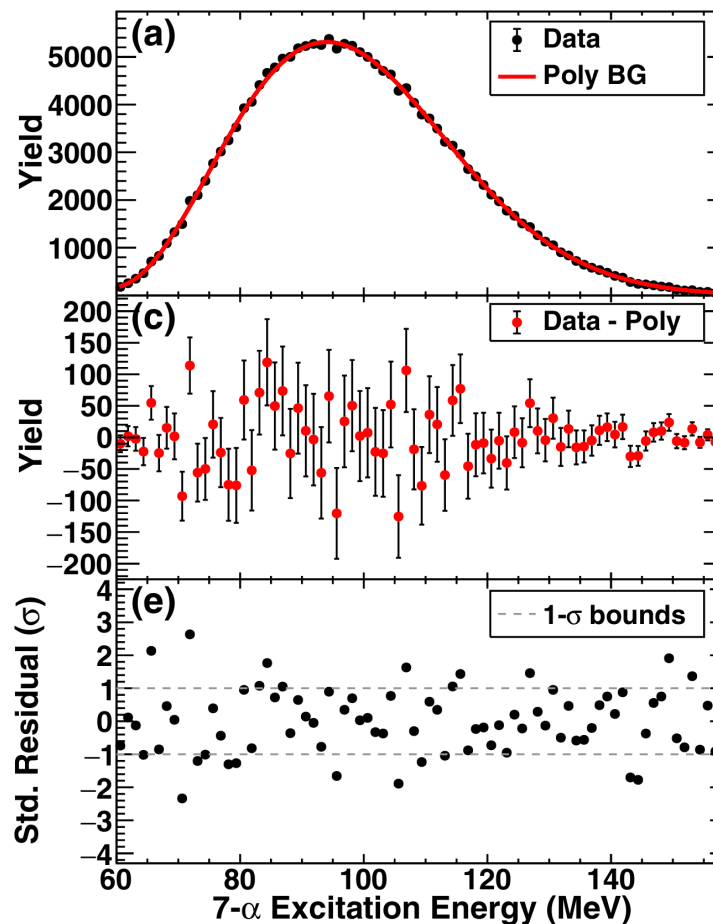
- Note, any broad distribution can also be fit with a linear combination of Gaussians, given enough terms.
- We explicitly allow for the possibility of very rare and wide features hiding in the statistical noise and background uncertainty
 - They would just need to be more prominent than the NIMROD analysis suggests



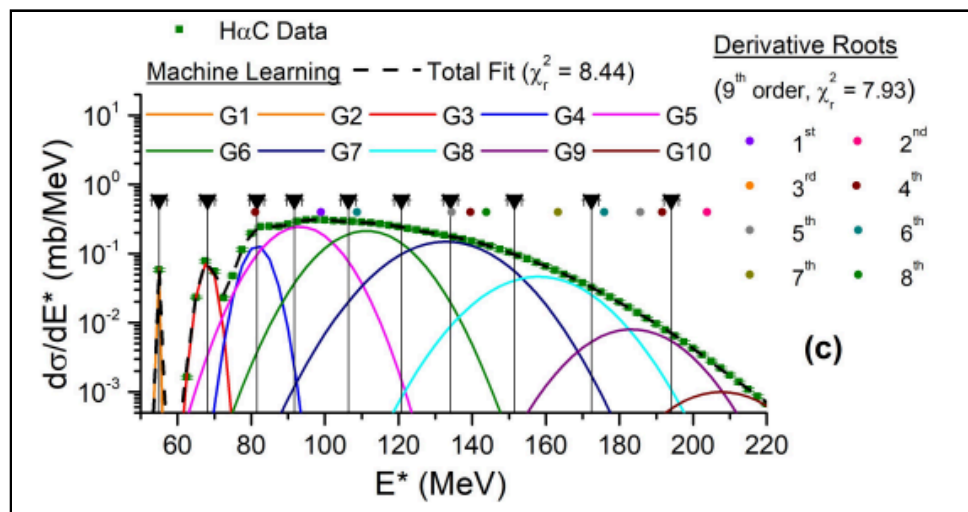
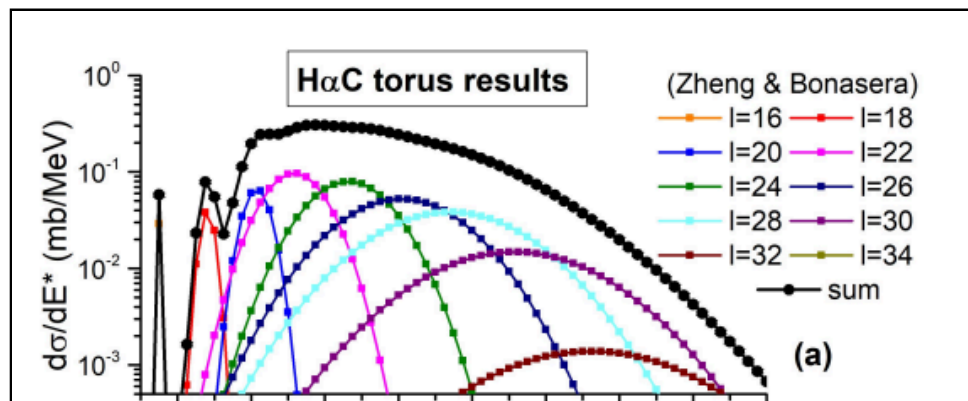
Polynomial Roots

“Second, a polynomial function possesses several ‘special’ points of maxima, minima and inflections that are given by the roots of each derivative of the polynomial. Since a polynomial of order N , can in principle have at most N roots, a subtractive analysis yields additional structure that is not present in the data. In our analysis, we obtain the “special” points numerically for each dataset, by fitting a 9th order polynomial and accept the roots on each derivative if they are real and their value of the next derivative is negative (2nd Derivative Criterion).” — §2 par 1

- There are not prominent fluctuations in the polynomial on the few MeV scale
- Checked derivatives and integral
- Also no structure in the std. residuals



Use of Polynomial Roots to Discredit the Polynomial Fit



“We observe that seven out of ten machine learning Gaussians reproduce the known peaks compared to four out of twelve polynomial roots. This demonstrates the superiority of the proposed AI method.”

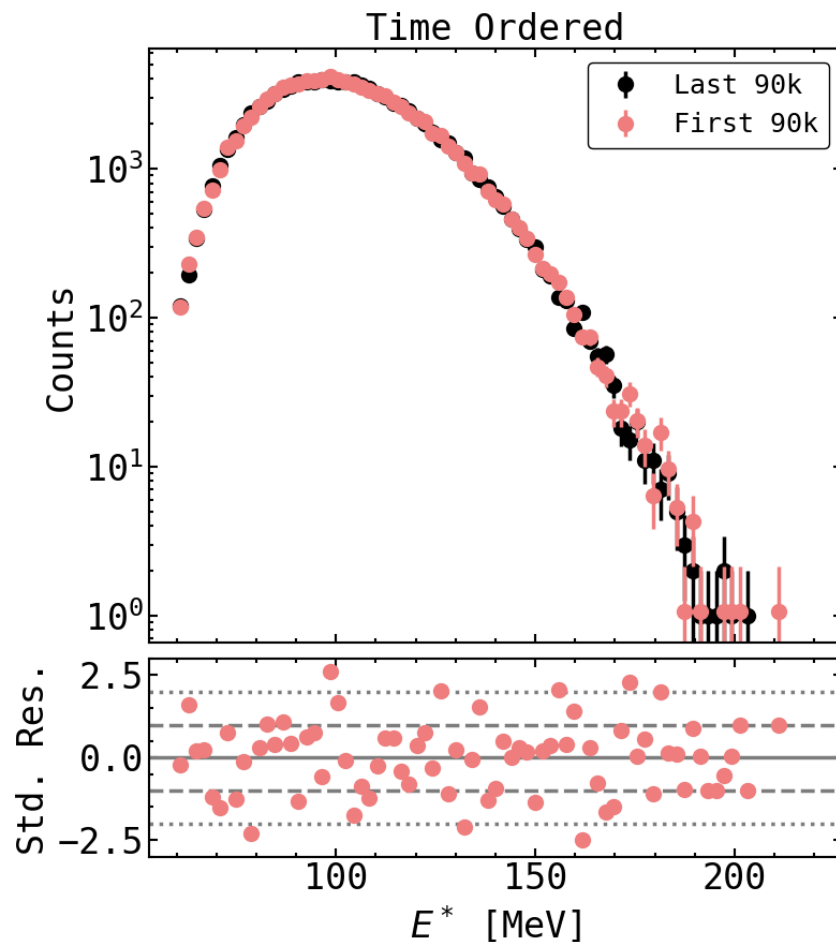
- The Hannaman analysis only used the polynomial to demonstrate a lack of structure
- The ‘special’ points were not used to predict underlying structure
- I agree using the ‘special’ points in this way is a bad idea

**There is not statistically significant
evidence for gain drifts in the time-
ordered $E_{7\alpha}^*$ spectrum**

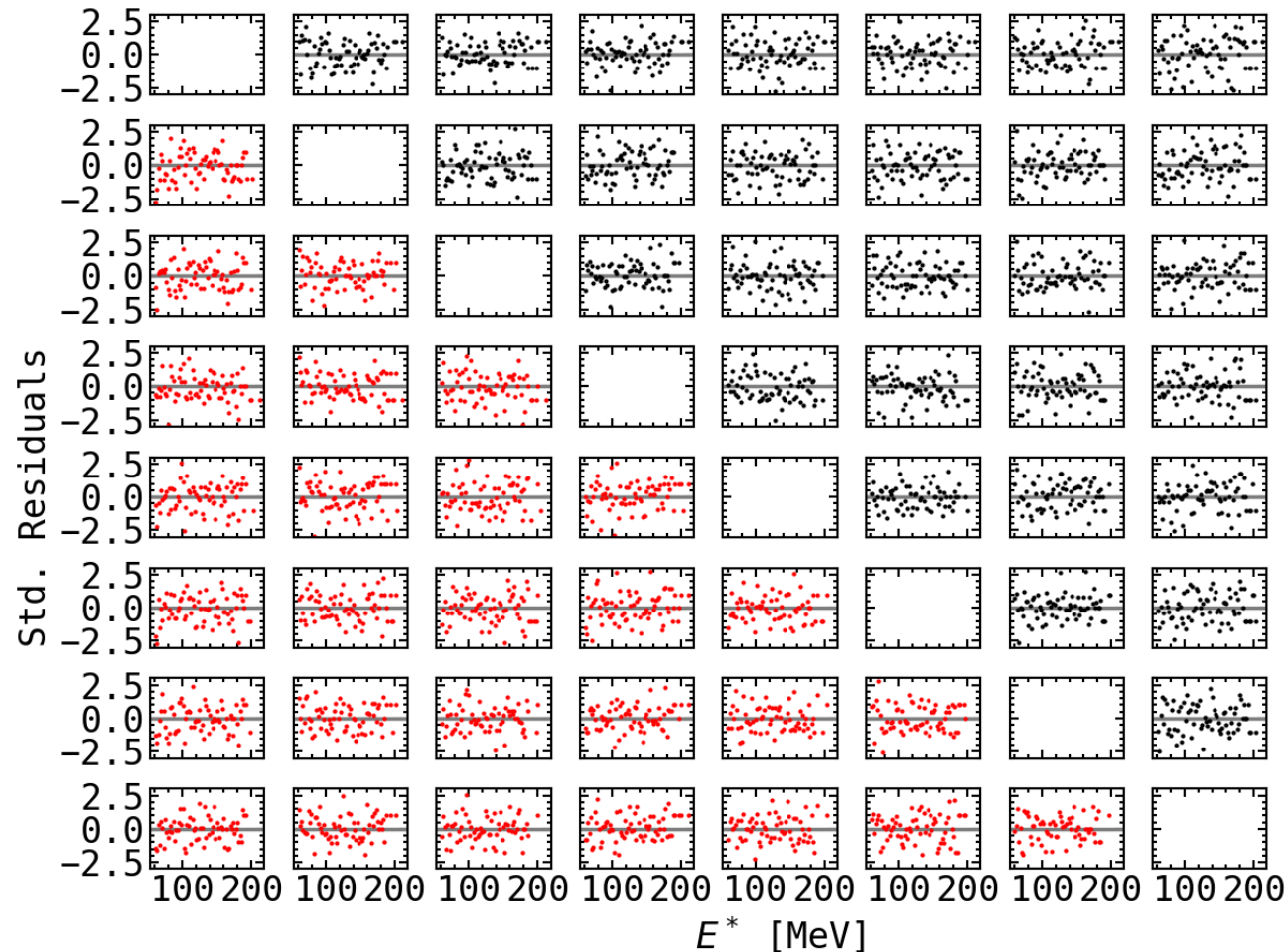
Looking for Gain Drifts

“We note that the excitation energies of each partition fluctuate around the full statistics value, with only variation of $\approx 2.6\%$. Their common tendencies might correspond to minor experimental shifts, which in our method are mostly compensated by the larger fluctuation of the widths.”

- No matter how you partition the time-ordered data, you always end up with statistically consistent spectra
- Right: a ~50-50 split for example
- Disclaimer: Treating Poisson as simple $\sqrt{N}$ due to timing constraints
 - (Careful looking too hard at the high E^* tail)

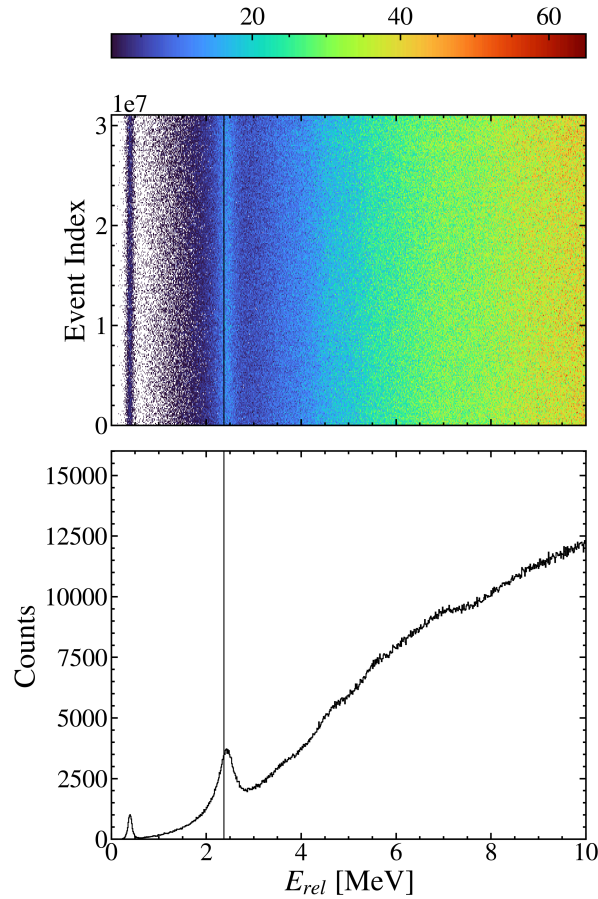
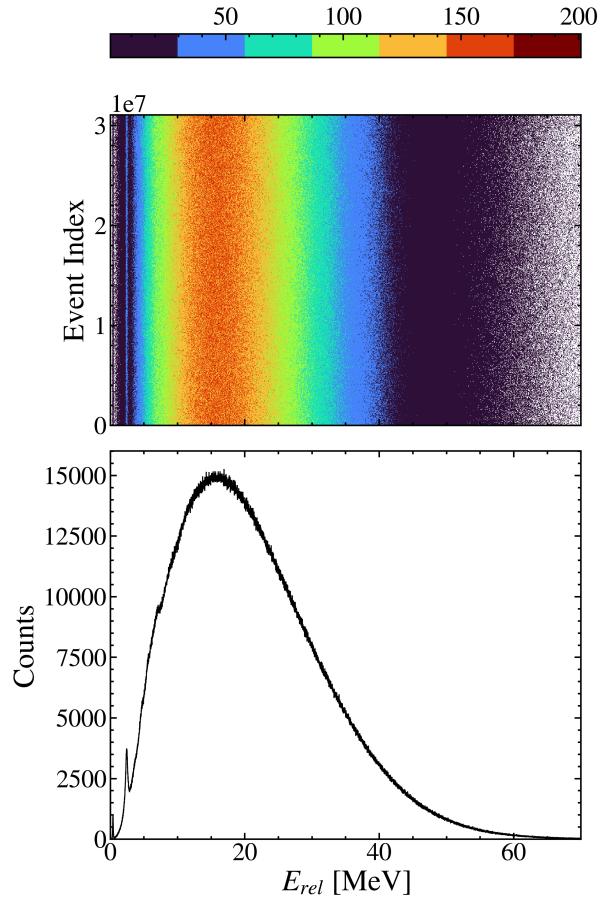


8-fold split



- **BLACK:** Time ordered
- **RED:** Time shuffled
- Variances of standardized residuals range from ~ 0.78 - ~ 1.4 for both
 - Typically between 0.85 and 1.15 for both

3α Spectra



- Time ordered event index versus $3\alpha-E_{rel}$
- $^{12}\text{C}(3^-)$ accurately reproduced
- There is no evidence for gain drifts

Summary

When is a peak consistent with a prediction?

The Cao candidate peaks are too far from the GMM peaks to be statistically significant

What is the background?

The possibility of the background must be accounted for

Use of Polynomial

Only used to show there is not strong narrow structure

Quality of Calibration

There are not detectable gain drifts

General Statistics

χ^2_V requires independent samples which supersampling breaks

Handling of experimental systematics

Acknowledgments



- SJY Group
- Andy Hannaman
- Department of Energy: DE-FG02-93ER40773



U.S. DEPARTMENT OF
ENERGY

Backup Slides

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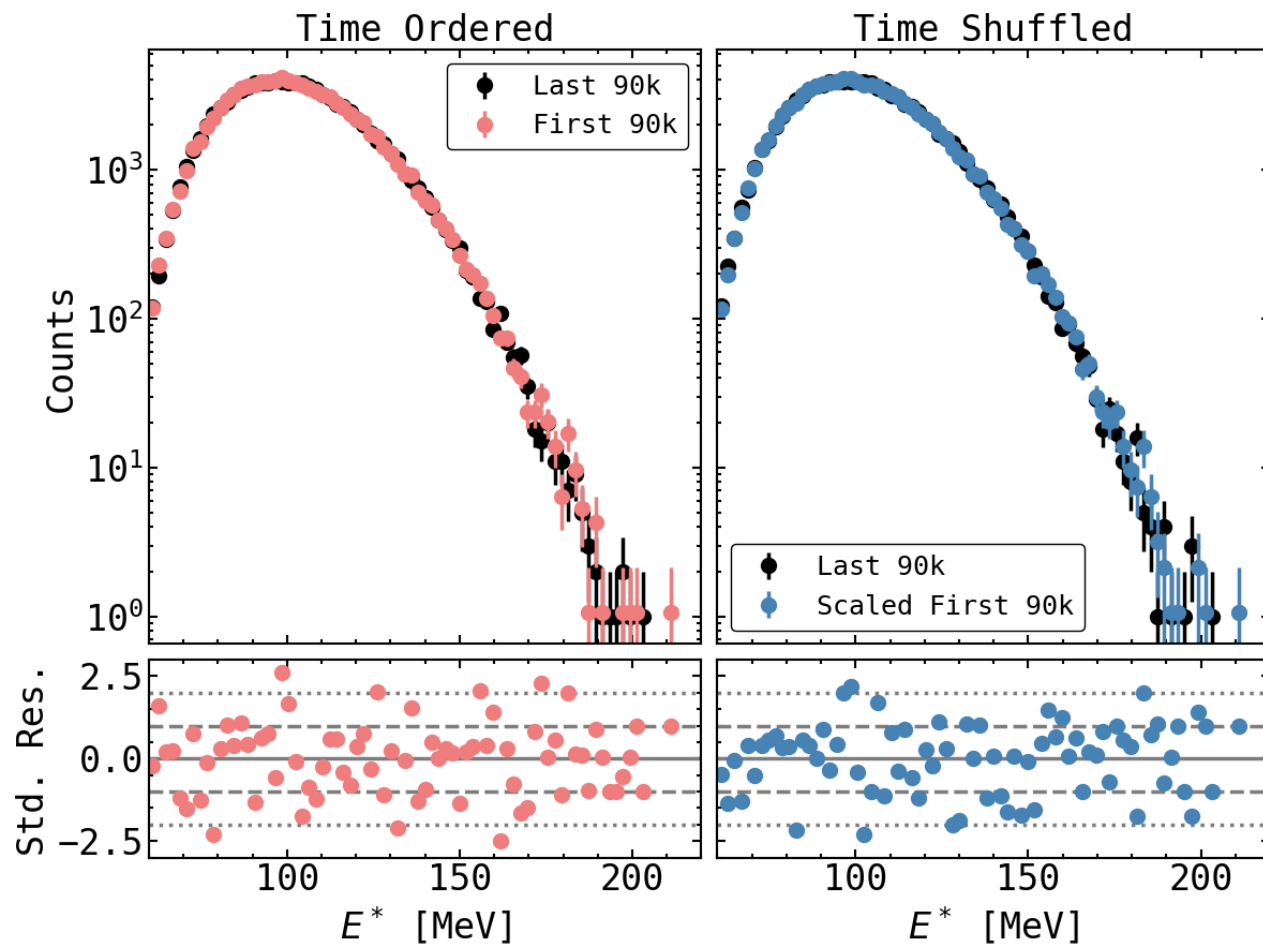
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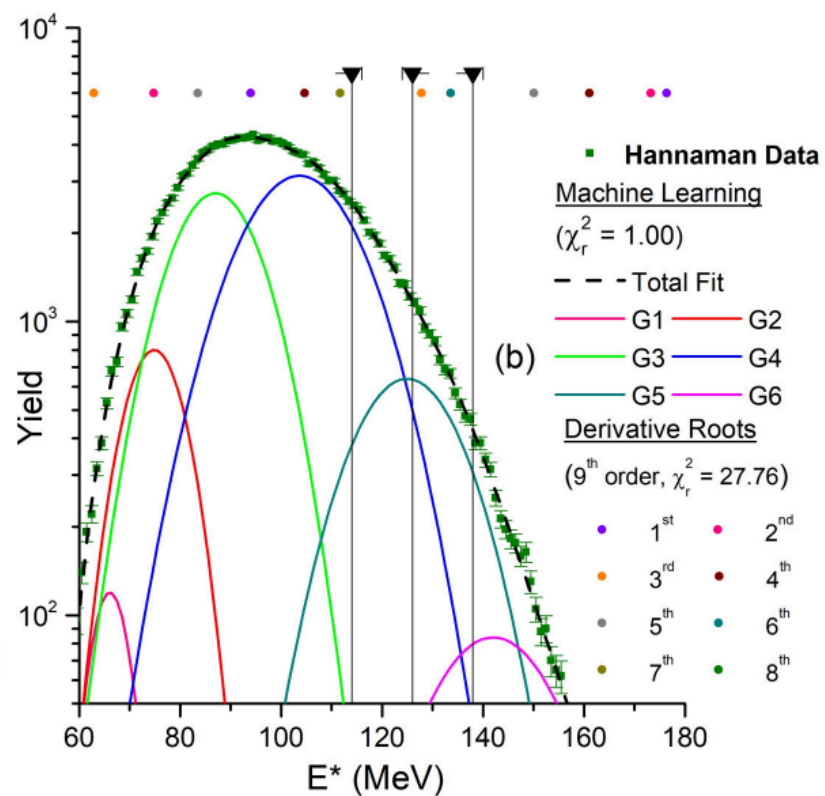
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With Shuffle

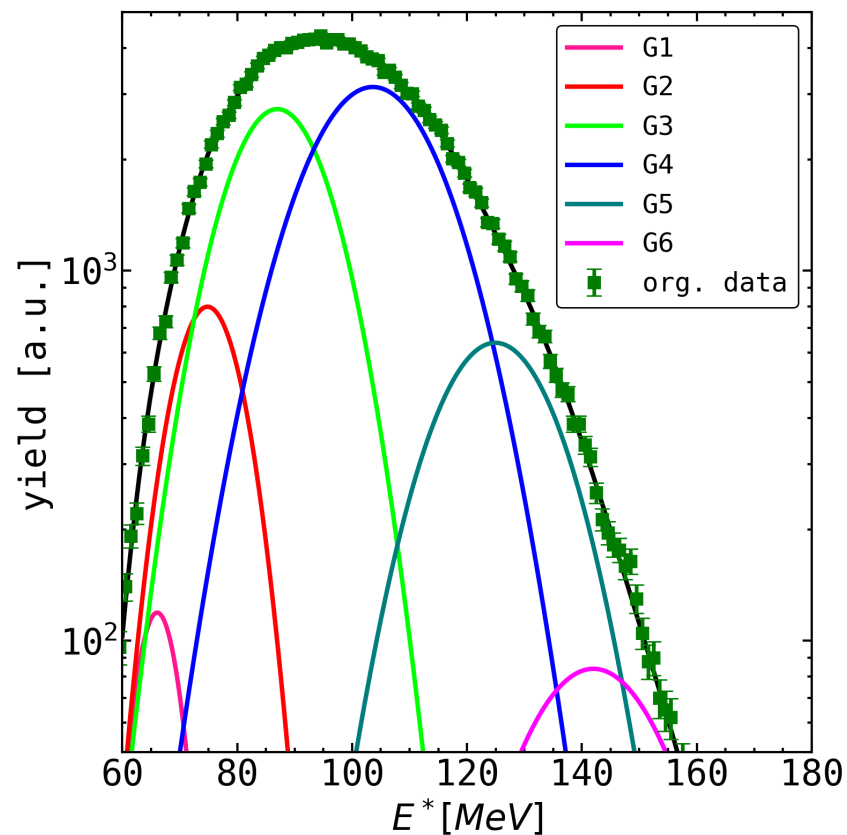


Reproduction of Paper data

Original Plot



Recreation Using Plot Digitizer



**Statistical Concerns with the GMM Technique as
Implemented
(And other miscellaneous concerns)**

Assigning 5% δy to model calculations

- **Spectrum is binned and bins are assigned uncertainty**

“For the calculation of χ_v^2 we assume a uniform 5% error” - §2 par. 4

- $\delta y_i = (0.05)y_i$

- *Binned data is sampled $N_p = 10^6$ times*

$$\chi_v^2 = \frac{\sum_i \left\{ \frac{1}{\delta y_i} \left[y_i - \sum_{l_G=0}^{N_G-1} w_{l_G} g_{l_G} \left(E_i^*; \mu_{l_G}, \sigma_{l_G} \right) \right] \right\}^2}{N_p - 3N_G}$$

- *Point cloud is passed to Gaussian Mixture Model for various N_G*

- This type of error does not reflect counting statistics

- **Select the model that optimizes χ_v^2**

- This might explain the degenerate peak ($G1 = G2$) in the benchmark

- *Locations of resultant Gaussians are compared to previous predictions*

- Probably need to sample the model 186k times and use the resultant spectra from that sampling with Poisson errors

Super Sampling

- *Spectrum is binned and bins are assigned uncertainty*

- **Binned data is sampled $N_p = 10^6$ times**

- *Point cloud is passed to Gaussian Mixture Model for various N_G*

- **Select the model that optimizes χ_v^2**

- *Locations of resultant Gaussians are compared to previous predictions*

“From this discrete histogram, we generate $N_p = 10^6$ random points, from a normal distribution around the center of each bin”

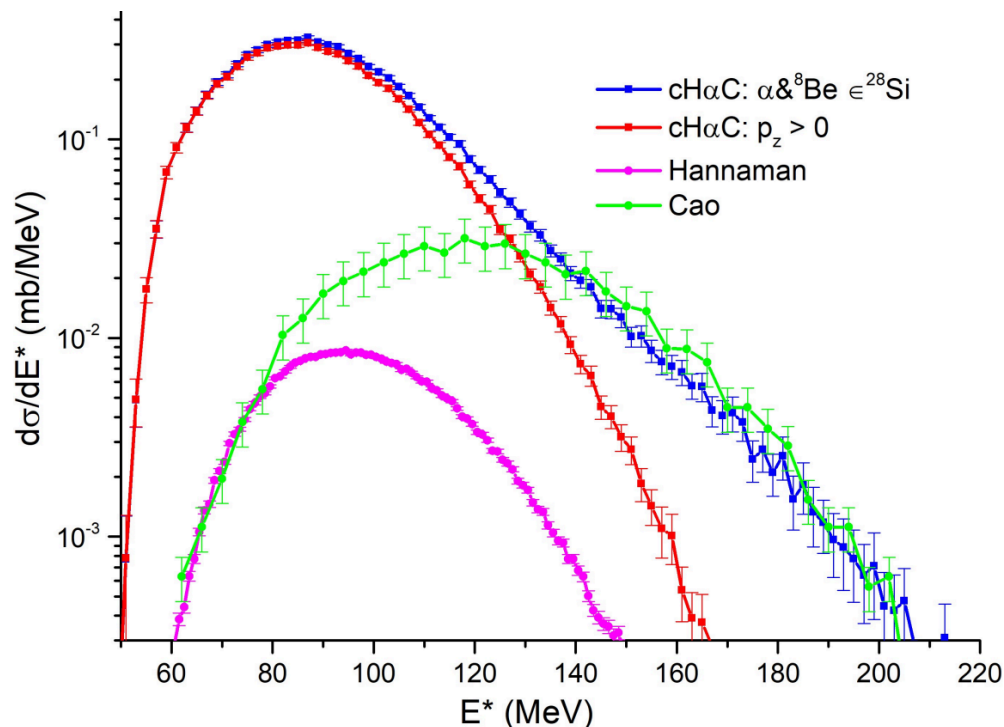
$$\chi_v^2 = \frac{\sum_i \left\{ \frac{1}{\delta y_i} \left[y_i - \sum_{l_G=0}^{N_G-1} w_{l_G} g_{l_G}(E_i^*; \mu_{l_G}, \sigma_{l_G}) \right] \right\}^2}{N_p - 3N_G}$$

- $N_p = 10^6$ samples from a condensed representation of the data (binning) are not independent samples
 - Especially when there are only 186k events in the most populated spectra
- This could create an unexpected bias on sample size as χ_v^2 should depend on N , but doesn't here

Super Sampling with Bin-Centered Gaussians

- *Spectrum is binned and bins are assigned uncertainty*
 - **Binned data is sampled $N_p = 10^6$ times**
 - *Point cloud is passed to Gaussian Mixture Model for various N_G*
 - *Select the model that optimizes χ_v^2*
 - *Locations of resultant Gaussians are compared to previous predictions*
- “From this discrete histogram, we generate $N_p = 10^6$ random points, from a normal distribution around the center of each bin”
- This by construction creates peaks at every bin center
 - Would not see effect without binning finer after supersampling
 - GMM takes the raw point cloud and can be affected
 - Unclear to me what the exact effect is

Comparison of Theory to Experiment w/o Experimental Response

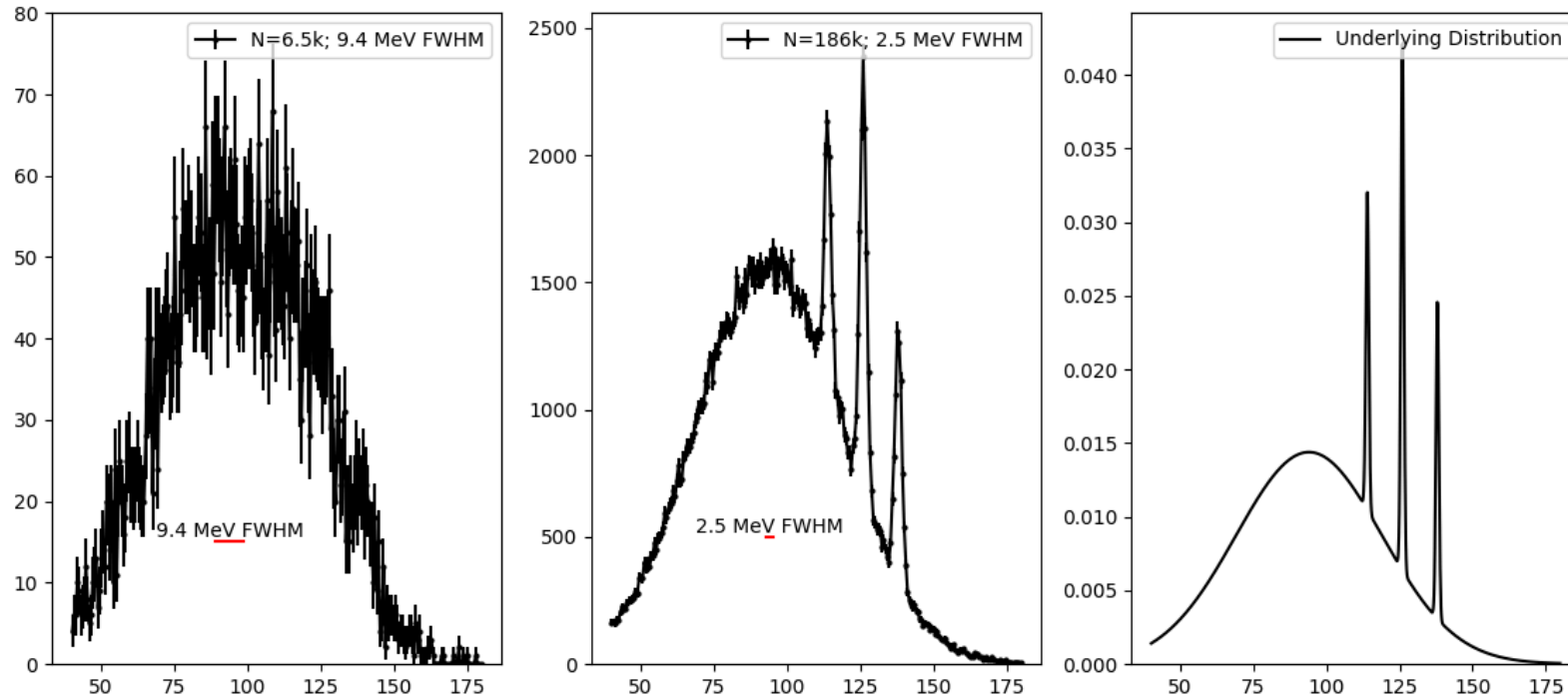


“We observe that the new cHαC calculation results converge to the experimental data in the high E^* tail, while they are generally higher at lower E^* values. ...

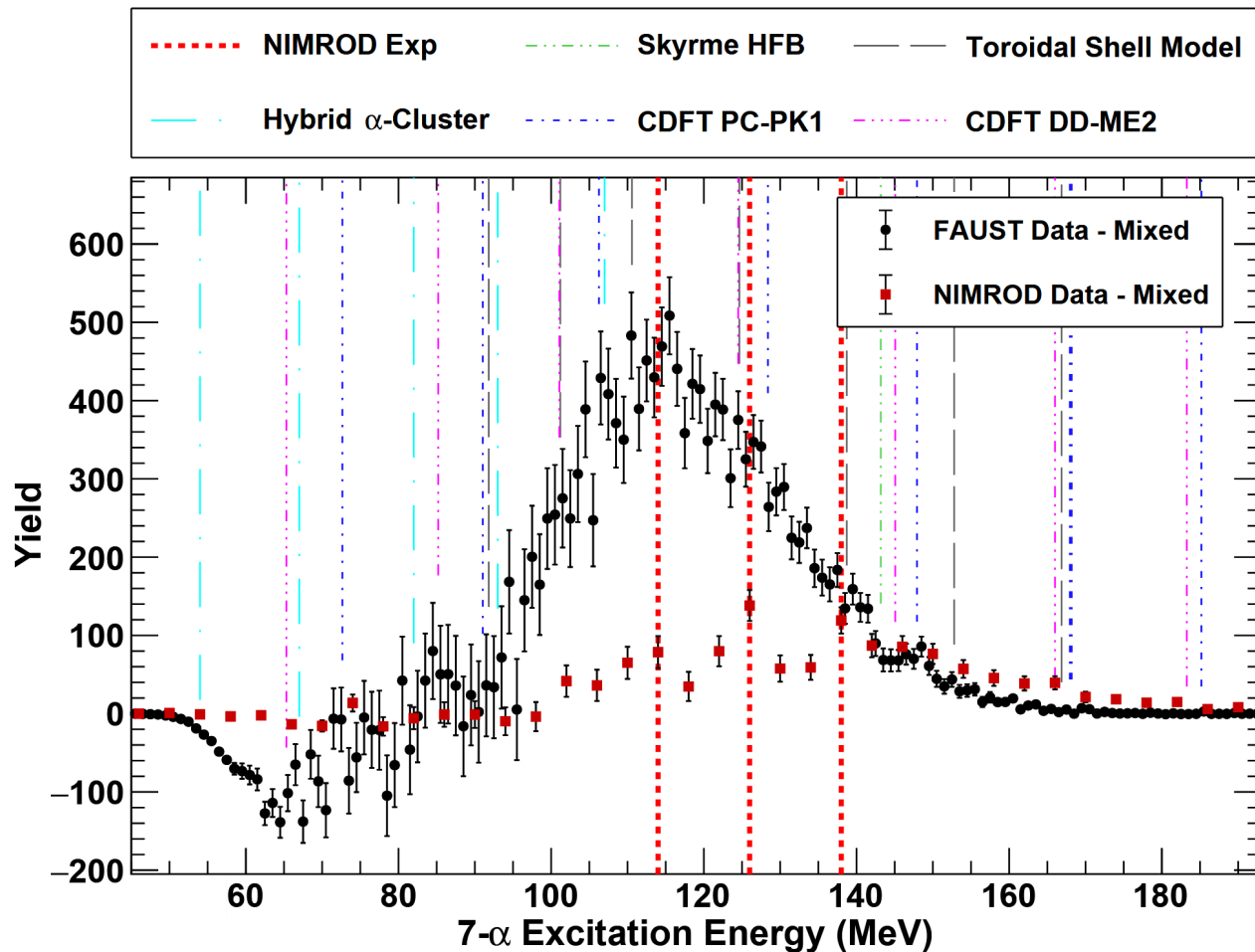
... On the contrary, for lower excitation energies, there are many open non- α -conjugate exit channels not available in our model.” — §3 par 7

- Simulations of FAUST and NIMROD responses were not used for this comparison.
- The (complex) effects of efficiency and resolution make direct comparisons dubious.

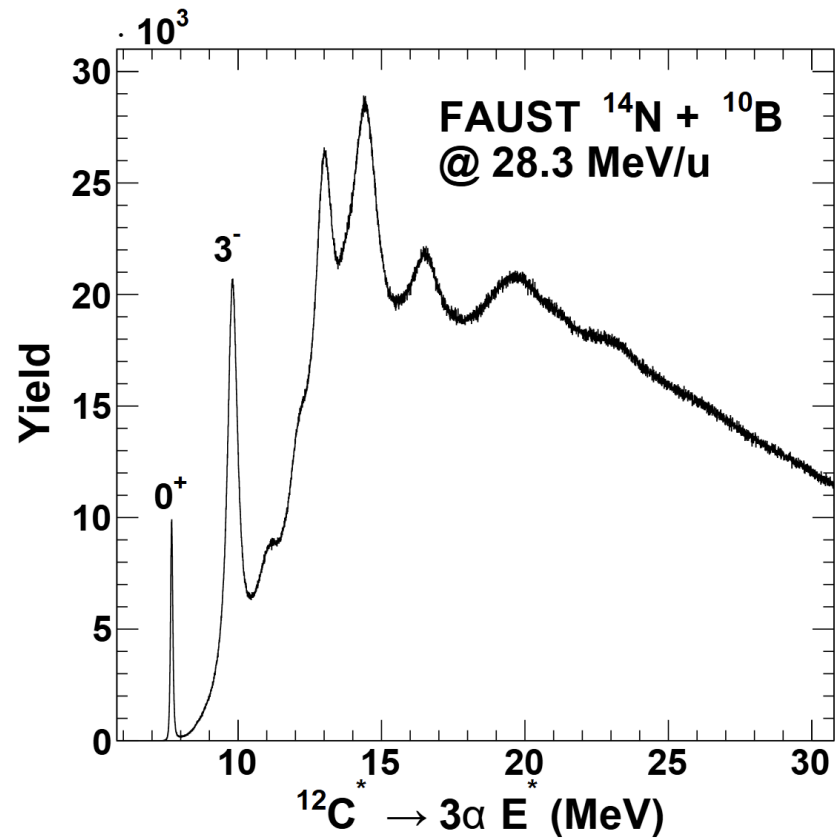
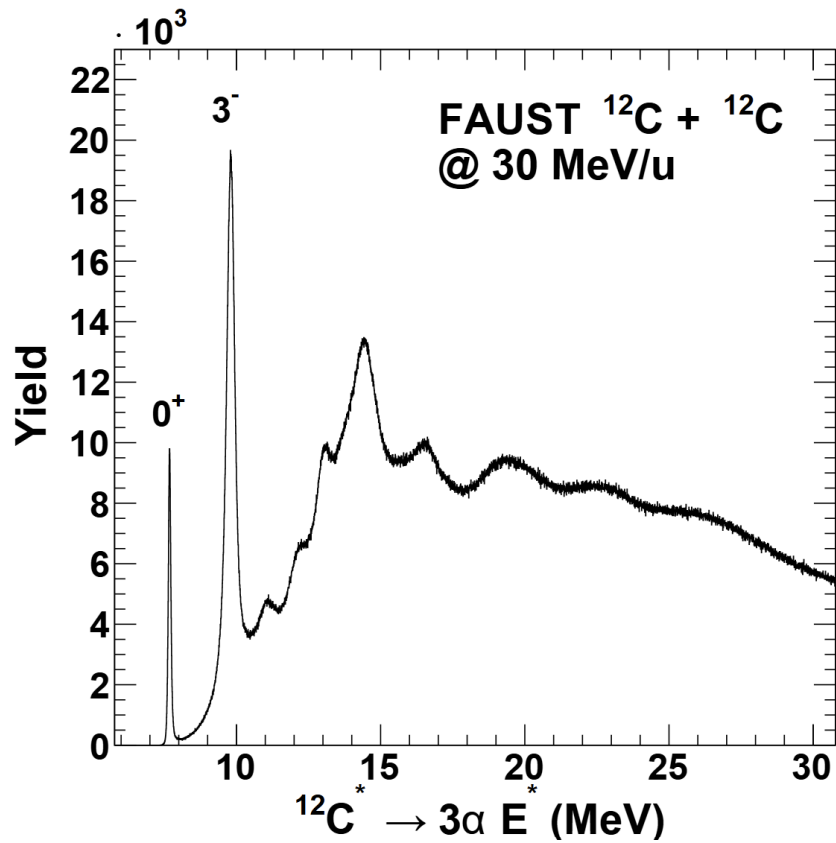
We Would See Prominent, Narrow Resonances



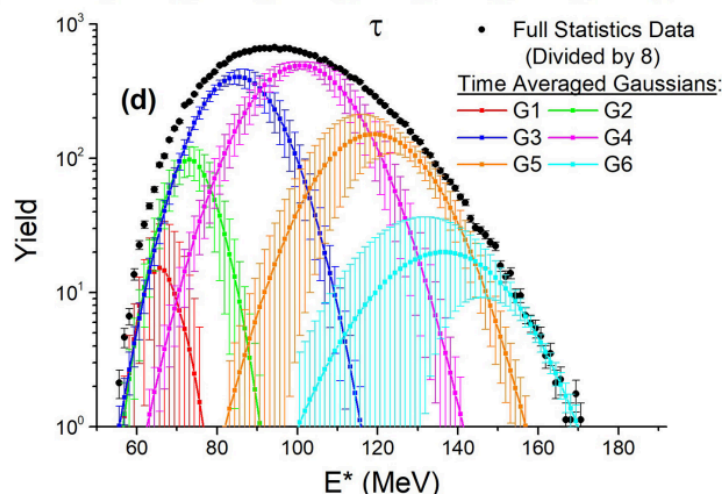
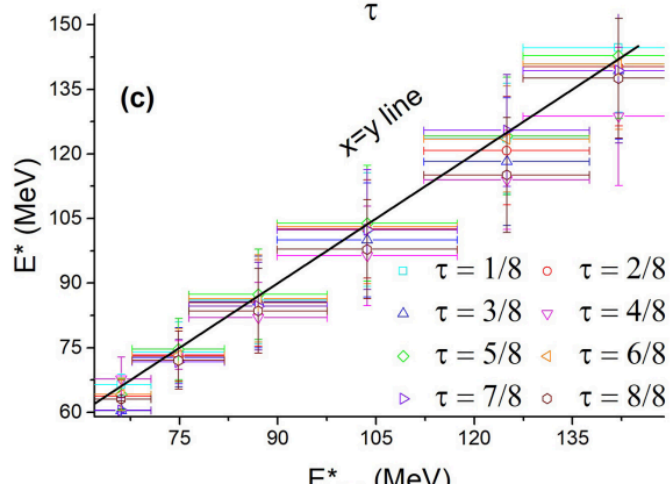
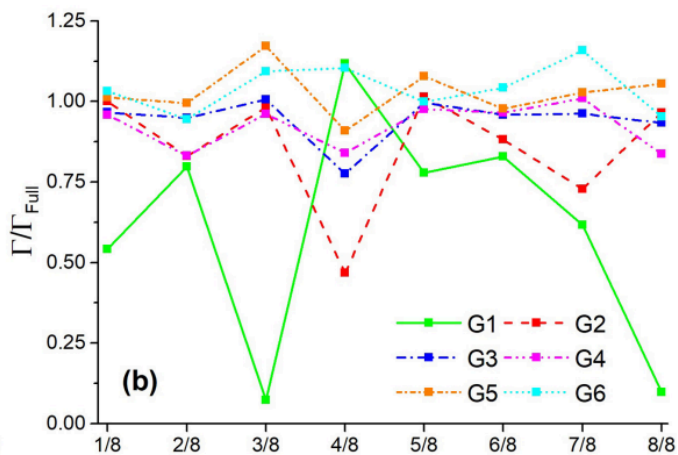
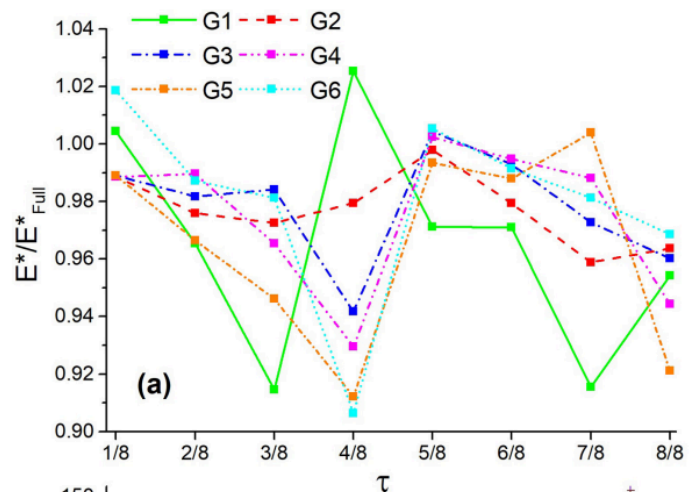
State of the Search as of 2023



Recent ^{12}C spectra



Partitioned Results



Partitioned Results

